

# Deep Belief Network Enabled Surrogate Modeling for Fast Preventive Control of Power System Transient Stability

Tong Su , Youbo Liu , *Member, IEEE*, Junbo Zhao , *Senior Member, IEEE*, and Junyong Liu, *Member, IEEE*

**Abstract**—The widely used transient stability-constrained optimal power flow (TSC-OPF) method for power system preventive control is very time-consuming and thus not applicable for large-scale systems. This article proposes a new deep learning-enabled surrogate model that can significantly improve computational efficiency while maintaining high accuracy. To achieve that, the deep belief network (DBN) is strategically integrated with the reference-point-based nondominated sorting genetic algorithm (NSGA-III) to develop a new preventive control framework. The DBN allows us to identify the mapping relationship between the transient stability index and system operational features. The identified functional mapping relationship is further used as the surrogate to connect the DBN results with TSC-OPF for preventive control. The integrated NSGA-III and surrogate model enable the multiobjective optimization to consider various constraints and objectives, such as minimization of costs of generation dispatch cost and load shedding while maintaining the system stability. Extensive simulation results on several IEEE test systems show that the proposed method can achieve highly efficient control solutions and outperform other alternatives in terms of computational efficiency and economic benefits.

**Index Terms**—Deep belief network (DBN), deep learning, nondominated sorting genetic algorithm (NSGA-III), surrogate model, transient stability preventive control.

## I. INTRODUCTION

WITH the increased penetration of renewable energy and scale of the power system, the system is operated more and more close to their stability limits. Therefore, the

Manuscript received December 7, 2020; revised March 6, 2021 and April 5, 2021; accepted April 8, 2021. Date of publication April 13, 2021; date of current version September 29, 2021. The work of Youbo Liu was supported in part by the National Natural Science Foundation of China under Grant 51977133. Paper no. TII-20-5525. (*Corresponding author: Youbo Liu.*)

Tong Su, Youbo Liu, and Junyong Liu are with the College of Electrical Engineering, Sichuan University, Chengdu 610065, China (e-mail: sutongscu@gmail.com; liuyoubo@scu.edu.cn; liujy@scu.edu.cn).

Junbo Zhao is with the Department of Electrical and Computer Engineering, Mississippi State University, Starkville, MS 39762 USA (e-mail: junbo@ece.msstate.edu).

Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TII.2021.3072594>.

Digital Object Identifier 10.1109/TII.2021.3072594

development of effective preventive controls for transient stability is of vital importance [1].

The traditional transient stability assessment usually relies on time-domain simulations. However, that is computationally intensive for practical large-scale systems [2]. To deal with that, several other methods have been proposed, including the phase-plane trajectories [3], the Lyapunov functions family [4], the ensemble Kalman filter [5], the unscented transformation [6], the transient energy function [7], and the nonparametric model [8]. Machine learning-based tools have also been applied for transient stability prediction. In [9], the improved decision tree is developed for online dynamic security assessment using synchrophasor measurements. It considers all possible paths instead of the classification results at each terminal node. Additionally, the Lasso regression is used to predict the transient stability boundary [10]. In [11], bus voltage magnitudes, the generator rotor speeds, and rotor angles are utilized as input parameters for the support vector machine (SVM) to predict transient stability. Instead of SVM, the artificial neural network (ANN) [12], [13] is used for that purpose as well. However, these traditional machine learning algorithms have rather limited capabilities of dealing with system nonlinearity and generalization. To overcome these issues, deep learning models are developed and applied in many fields. Compared to the traditional machine learning techniques, deep learning has the advantages of automatic feature extraction, strong abstraction ability, and good convergence. Particularly, the deep neural network structure has a larger number of neural layers. This allows uncovering the underlying knowledge contained in a large set of data. In [14], for instance, the stacked denoising autoencoder is used for transient stability prediction. However, it does not answer the question of how to leverage the predicted results for transient stability control. This article aims to bridge this gap and develop a new deep learning-based surrogate model for transient stability preventive control.

Preventive control aims to ensure the reliable operation of power systems in a future horizon, e.g., the next 15 min, by triggering appropriate actions in advance, such as generation redispatch, load shedding, and topology reconfiguration. The preventive control is derived from the optimal power flow (OPF) model with several constraints considering predefined contingencies. Transient stability-constrained OPF (TSC-OPF) is one of the most widely used approaches to determine preventive

actions with consideration of stability margins in the presence of multiple faults [15]. However, TSC-OPF is computational-prohibitive for online preventive control. To deal with that, in [16], the trajectory sensitivity is leveraged on the one-machine infinite bus system. But the trajectory sensitivity could not handle multiobjective optimization problems. In [17], the dynamic secure/insecure regions of the system are determined through pattern recognition for preventive dynamic security control. However, the key features are manually extracted, which may yield degraded performance for more complicated scenarios. It is worth pointing out that very few machine learning models [18] have been used for preventive control in the literature and they are computational demanding for practical applications [19], or only for single-objective optimization without considering the cost and other power systems operational status [20]–[23]. This restricts their applicability of dealing with multiobjectives, such as balancing the minimization of generator dispatch costs and minimizing load shedding.

In this article, a new deep-learning-enabled computationally efficient transient stability preventive control framework is proposed. It has the following two main contributions.

- 1) The deep belief network (DBN) is strategically integrated with the reference-point-based nondominated sorting genetic algorithm (NSGA-III) to develop the new multi-objective optimization framework for preventive control. This allows us for the first time to connect the transient stability predictions from deep learning with the multi-objective optimizations, including minimizing the costs of generator redispatch and load shedding. Specifically, DBN maps the relationship between power system operational features and transient stability margin constrained by predefined contingencies. This leads to the development of a surrogate model for physical transient stability constraints. The latter are traditionally modeled by a large set of differential-algebraic equations and are the main computing bottleneck. The surrogate model significantly reduces the computational complexity of the full TSC-OPF by simply performing direct neural network-based transient stability verification in each iterative searching step for the optimal preventive control strategy, rather than executing many time-domain simulations.
- 2) The traditional approaches, such as the interior point method and the convex programming, can hardly address the optimization problem with implicit black-box constraints derived from neural networks [24]. To deal with black-box constraints, the NSGA-III algorithm that can find the Pareto-optimal front of multiobjective is integrated with the DBN enabled the surrogate model to coordinate the multiple objectives. Comparative results with other methods demonstrate that the new framework achieves significantly improved computational efficiency without loss of control performance.

The rest of this article is organized as follows. Section II presents the DBN-based surrogate modeling methodology. Section III shows the proposed framework. Numerical results are presented and analyzed in Section IV. Finally, Section V concludes this article.

## II. SURROGATE MODELING METHODOLOGY

### A. Generation of Data Samples

Preventive control aims to change the system operating point by shifting generator outputs and/or load shedding to avoid the risk of violating security criteria. Note that the knowledge of generator voltage magnitudes and real power injections, as well as the real power injections and real and reactive power injections of all loads, is sufficient to determine the initial conditions of all generators for transient stability simulations. The latter determines the transient stability index (TSI). Therefore, the operational features  $\Lambda = (\mathbf{P}_G, \mathbf{V}_G, \mathbf{P}_L, \mathbf{Q}_L)^T$  that include the voltage magnitudes of the generators  $\mathbf{V}_G$ , the active power of the generators  $\mathbf{P}_G$  and loads  $\mathbf{P}_L$ , and reactive power of the loads  $\mathbf{Q}_L$  are selected as the input features for DBN. To achieve high accuracy and good generalization, the data samples for training DBN should contain all possible operational features. This means that the data samples should be representative and evenly distributed in the sampling space. In this article, the Latin hypercube sampling (LHS) is applied to generate data samples associated with each operation scenario at the same time.

LHS is a multidimensionally distributed statistical method that generates random samples of the focused variables. Latin square (LS) refers to a square grid with one sample in each row and column, and Latin hypercube (LH) is the generalization of LS in the multidimensional scenario. Let  $N_S$  denote the number of samples, LHS divides each dimension of sample space into  $N_S$  equal parts and then gets sampling according to the LH principle. This sampling method ensures the independence and uniformity of data samples.

Besides varying generation-load balance in the sampling process, the permanent three-phase short-circuit faults on given lines are considered as pre-contingencies for each scenario. Through extensive off-line time-domain simulations, we can obtain generator rotor angle trajectories and transient stability index, TSI. TSI reflects the maximum rotor angle difference of the generators in the transient process and its mathematical expression is [17]

$$\text{TSI} = \frac{360 - |\delta_{\max}|}{360 + |\delta_{\max}|} \times 100 \quad (1)$$

$$\delta_{\max} = \max_{t \in T} (\delta_{i,t} - \delta_{j,t}) \quad (2)$$

where  $\delta_{\max}$  is the maximum rotor angle difference between any two generators; and  $T$  is the observation interval. When  $\text{TSI} > 0$ , the system is transient stable while  $\text{TSI} < 0$  means system unstable. Note that the increase of TSI reflects transient stability enhancement and vice versa.

Assuming that the number of samples for the operational features generated by the LHS is  $k$  and the number of contingencies is  $p$ , then  $k \times p$  time-domain simulations are performed. To ensure that DBN can predict the TSI corresponding to the most severe scenario derived from the predefined contingencies, the operational features and the minimum TSI of  $p$  TSIs are selected to form the sample sets for DBN training. It should be clarified that the scope of the article is a preventive control. It is well known that preventive control aims to deal with severe

contingencies with high probability and thus could not deal with all possible contingencies. If preventive controls are not effective in addressing some unexpected events, emergent controls are needed, which is out of the scope of this article.

### B. Deep Belief Network-Based Transient Stability Predictor

DBN is a probability generation model, which is composed of a multilayer restricted Boltzmann machine (RBM) and a fully connected layer. RBM is a two-layer unsupervised network that contains a visible layer and a hidden layer. The main feature of RBM is its ability to probabilistically reconstruct input features by two-way connections.

DBN learning has two main steps: pretraining and fine-tuning. Pretraining solves the problems of gradient disappearance and the explosion of the deep neural network. In the pretraining step, RBM is trained layer by layer through the input data. Then, DBN reconstructs multiple features of input data. In the fine-tuning step, DBN further trains the model with supervision to fit input data and labels.

RBM is an energy-based model. Defining  $v$  and  $h$  the visible layer and the hidden layer matrices, respectively, the energy of a configuration  $(v, h)$  is

$$E(v, h) = -a^T v - b^T h - v^T W h \quad (3)$$

where  $a$  and  $b$  are the bias matrices of the visible layer and the hidden layer; and  $W$  is the weight matrix between the visible layer and the hidden layer. The probability distributions of  $v$  and  $h$  can be obtained from the energy of  $(v, h)$

$$P(v, h) = \frac{1}{Z} e^{-E(v, h)} \quad (4)$$

$$Z = \sum_v \sum_h e^{-E(v, h)} \quad (5)$$

where  $Z$  is the partition function, also known as the normalizing constant, the aim of which is to ensure the sum of the probability distribution is 1.

From (3), the marginal probability of the visible layer vector can be calculated by accumulating all possible hidden layer configurations via

$$P(v) = \frac{1}{Z} \sum_h e^{-E(v, h)}. \quad (6)$$

Since RBM lacks intralayer connections, the activations of the visible units and hidden units are independent. In other words, given the configuration of the hidden unit  $h$ , the conditional probability of the visible unit  $v$  is

$$P(v|h) = \prod_{i=1}^m P(v_i|h). \quad (7)$$

Similarly, the conditional probability of  $h$  given  $v$  is

$$P(h|v) = \prod_{j=1}^n P(h_j|v). \quad (8)$$

The individual activation probabilities are given by

$$P(h_j = 1|v) = \text{SeLU} \left( \sum_{i=1}^m w_{i,j} v_i + b_j \right) \quad (9)$$

$$P(v_i = 1|h) = \text{SeLU} \left( \sum_{j=1}^n w_{i,j} h_j + a_i \right) \quad (10)$$

$$\text{SeLU} = \begin{cases} x, & x \geq 0 \\ \alpha(e^x - 1), & x < 0 \end{cases} \quad (11)$$

where SeLU is the activation function and  $\alpha$  is the scale factor.

Training RBM by maximizing the expected log probability of a few training sets  $V$  yields

$$\arg \max_W \prod_{v \in V} P(v). \quad (12)$$

Taking  $w$  as an example, the partial derivation of  $w$  can be obtained as follows:

$$\frac{\partial \ln(P(v))}{\partial w_{i,j}} = P(h_j = 1|v) v_i^t - \sum_v P(v) P(h_j = 1|v) v_i. \quad (13)$$

The weight  $w_{i,j}$  is updated by

$$w_{i,j} = w_{i,j} + \eta \frac{\partial \ln(P(v))}{\partial w_{i,j}} \quad (14)$$

where  $\eta$  denotes the learning rate.

Similarly, the energy function and probability distribution function of DBN are as follows:

$$E(v, h^1, h^2, h^3) = -v^T W^1 h^1 - h^{1T} W^2 h^2 - h^{2T} W^3 h^3 \quad (15)$$

$$P(v, h^1, h^2, h^3) = \frac{1}{Z} e^{-E(v, h^1, h^2, h^3)}. \quad (16)$$

In this article, the TensorFlow framework is used to build the DBN model. The operational features are taken as input vectors, and the TSIs are used as the corresponding labels to learn the nonlinear mapping relationship between them. After training, TSI can be obtained quickly through transient stability predictor, which can be expressed as follows:

$$\text{TSI}_P = \mathcal{L}(\Lambda) \quad (17)$$

where  $\mathcal{L}(\Lambda)$  is the transient stability predictor and its input includes the operational features while its output is the TSI prediction, called  $\text{TSI}_P$ .

When training the DBN-based predictor, normalization is used to regularize data of different dimensions. The learning rate decay algorithm is used to speed up the learning speed and accuracy. In addition,  $L_2$  regularization is used to prevent model overfitting. Their mathematical expressions are

$$F_{i,N} = \frac{F_i - F_{i,\min}}{F_{i,\max} - F_{i,\min}} \quad (18)$$

$$\eta = \eta_0 \times \frac{1}{1 + D \times E} \quad (19)$$

$$LF = \frac{1}{2r} \sum_{\Lambda T} \|\mathcal{L}(\Lambda_T) - \text{TSI}_T\|^2 + \frac{\lambda}{2r} \sum_w w^2 \quad (20)$$

where  $F_i$  and  $F_{i,N}$  denote, respectively, the original and normalized data of the  $i$ th dimension data;  $F_{i,\max}$  and  $F_{i,\min}$  are the maximal and minimal values of  $F_i$ ;  $\eta$  is the learning rate;  $\eta_0$  is the initial value of  $\eta$ ;  $D$  is the learning rate attenuation coefficient;  $E$  is epoch numbers;  $LF$  is the loss function with  $L_2$  regularization, in which the first part is the mean square error (MSE) and the second part is the L2 regularization loss;  $r$  is the size of the training set;  $\Lambda_T$  is the operational features of the test set;  $TSI_T$  is the TSI of the test set; and  $\lambda$  is a regularization parameter.

In this article, the accuracy of the predictor is calculated by the following equation:

$$\begin{cases} A_{PC} = 1 - \frac{|(TSI_T < 0) \wedge (TSI_P > TSI_B)|}{N_T} \\ A_{DBN} = \frac{|\text{SIGN}(TSI_T) \leftrightarrow \text{SIGN}(TSI_P)|}{N_T} \end{cases} \quad (21)$$

where  $A_{PC}$  is the prediction accuracy of the preventive control;  $\wedge$  is the logical conjunction;  $N_T$  is the size of the test set; note that the second part after the negative sign represents the proportion that the scenarios predicted to be stable but should be unstable;  $A_{DBN}$  is the prediction accuracy of DBN;  $\leftrightarrow$  is the logical biconditional;  $\text{SIGN}$  is a function to find the sign of a value; and  $TSI_B$  is the critical TSI to assess the transient stability of power system.

### C. Surrogate-Assisted TSC-OPF

Traditional preventive control relies on the TSC-OPF model with predefined contingency constraints. The aim is to take appropriate actions to change the power system operation point while achieving the optimal objective subject to all constraints. However, the transient stability constraints consist of a set of differential-algebraic equations, which are computationally extensive to assess for online applications. To address that, the DBN-based predictor, for the first time, is introduced to surrogate the most computationally complicated parts of the TSC-OPF model. The surrogate model is much more computationally efficient than the original time-domain simulations. Specifically, the surrogate-assisted preventive control model is presented as follows.

**1) Objective Function:** It aims to minimize the cost and load ratio of system components, especially generators and transmission lines.

$$F_{\text{cost},t} = \sum_{i \in S_g} (C_{Ui} |\Delta P_{Ui,t}| + C_{Di} |\Delta P_{Di,t}|) + \sum_{j \in S_{ls}} (C_{Lj} |\Delta P_{Lj,t}|) \quad (22)$$

$$R_{a,t} = \frac{1}{M} \sum_{k=1}^M \frac{|P_{RLk,t}|}{P_{RCk,t}} \times 100\% \quad (23)$$

$$\Delta P_{Ui,t} = \begin{cases} P_{Gi,t} - P_{Gi,t-1}, & P_{Gi,t} > P_{Gi,t-1} \\ 0, & P_{Gi,t} \leq P_{Gi,t-1} \end{cases} \quad (24)$$

$$\Delta P_{Di,t} = \begin{cases} 0, & P_{Gi,t} \geq P_{Gi,t-1} \\ P_{Gi,t} - P_{Gi,t-1}, & P_{Gi,t} < P_{Gi,t-1} \end{cases} \quad (25)$$

$$\Delta P_{Lj,t} = P_{Lj,t-1} - P_{Lj,t} \quad (26)$$

where  $t$  represents time instant;  $i$  is the generator number;  $j$  is the load number;  $k$  is the component number;  $M$  is the number of components;  $F_{\text{cost},t}$  is the total cost;  $C_{Ui}$  and  $C_{Di}$  are the costs of increasing and decreasing the generator outputs;  $C_{Lj}$  is the cost of load shedding;  $\Delta P_{Ui,t}$  and  $\Delta P_{Di,t}$  are the sizes of increasing and decreasing the generator outputs;  $\Delta P_{Lj,t}$  is the size of the load shedding;  $P_{RLk,t}$  is the current load of component  $i$ ;  $P_{RCk,t}$  is the capacity of component  $i$ ;  $R_{a,t}$  is the average load ratio of components, including the average load ratio of generators and transmission lines;  $S_{ls}$  denotes the set of loads that can be shed; and  $S_g$  denotes the generator set.

**2) Equality Constraints:** Power flow equations:

$$\begin{aligned} P_{Gi,t} &= P_{Li,t} + \sum_{j=1}^n |V_{i,t}| |V_{j,t}| (G_{ij} \cos \alpha_{ij,t} + B_{ij} \sin \alpha_{ij,t}) \\ Q_{Gi,t} &= Q_{Li,t} + \sum_{j=1}^n |V_{i,t}| |V_{j,t}| (G_{ij} \sin \alpha_{ij,t} - B_{ij} \cos \alpha_{ij,t}) \end{aligned} \quad (27)$$

$i, j \in S_n$

where  $P_{Gi}$  and  $Q_{Gi}$  are active and reactive power generations at bus  $i$ ;  $P_{Li}$  and  $Q_{Li}$  are active and reactive load injections at bus  $i$ ;  $|V_i|$  and  $|V_j|$  are voltage magnitudes at bus  $i$  and  $j$ ;  $\alpha_{ij}$  is bus voltage angle difference;  $G_{ij}$  and  $B_{ij}$  are the real and imaginary parts of the line admittance; and  $S_n$  denotes the bus set.

**3) Inequality Constraints:** Operational constraints:

$$\begin{cases} P_{Gi}^{\min} \leq P_{Gi,t} \leq P_{Gi}^{\max}, i \in S_g \\ Q_{Gi}^{\min} \leq Q_{Gi,t} \leq Q_{Gi}^{\max}, i \in S_g \\ V_i^{\min} \leq V_{i,t} \leq V_i^{\max}, i \in S_n \\ I_{ij}^{\min} \leq I_{ij,t} \leq I_{ij}^{\max}, (i,j) \in S_l \\ -R_{Di} < (P_{Gi,t} - P_{Gi,t-1})/\Delta t < R_{Ui}, i \in S_g \end{cases} \quad (28)$$

where  $P_{Gi}^{\max}$  and  $P_{Gi}^{\min}$  are the upper and lower limits of generation;  $Q_{Gi}^{\max}$  and  $Q_{Gi}^{\min}$  are the upper and lower limits of the reactive power outputs;  $V_i^{\max}$  and  $V_i^{\min}$  are the upper and lower limits of bus voltages;  $I_{ij}^{\max}$  and  $I_{ij}^{\min}$  are the boundaries of thermal limits of line  $ij$ ;  $S_l$  denotes the line set;  $R_U$  and  $R_D$  are ramp ratios corresponding to increasing and decreasing generator outputs; and  $\Delta t$  is preventive control interval.

**4) Transient Stability Constraint:** Traditional transient stability constraints are  $|\delta_{i,t} - \delta_{i,t}^{COI}| \leq \delta^{\max}$ ,  $i \in S_c$  obtained from time-domain simulations and are computationally prohibitive. To deal with that, it is surrogated by the following proposed computationally efficient transient predictor:

$$\mathcal{L}(\Lambda_t) > TSI_B \quad (29)$$

where  $\mathcal{L}(\cdot)$  is the DBN-based surrogate operator. When  $TSI > TSI_B$ , the system is transient stable for the given operational features and vice versa.  $\mathcal{L}(\cdot)$  provides the fast-enough transient stability awareness in every searching iteration close to the optimal preventive control actions. The proposed DBN-based surrogate model for TSC-OPF is shown in Fig. 1.

## III. SOLUTION METHODOLOGY

The traditional approaches, such as the interior point method and the convex programming, could not address the formulation with implicit black-box constraints [24], such as the surrogate model in this article. Meanwhile, the widely used evolutionary



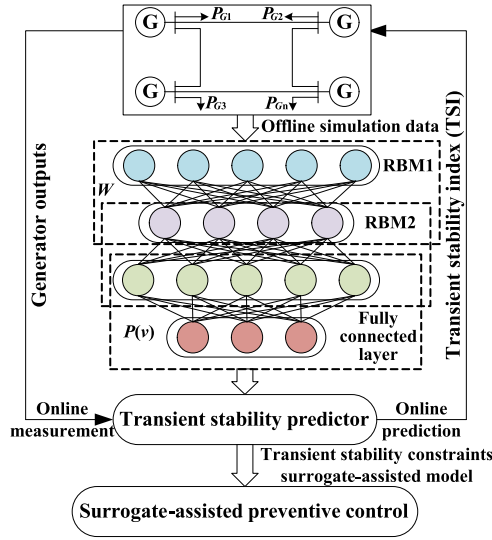


Fig. 1. DBN based surrogate model for multiobjective TSC-OPF.

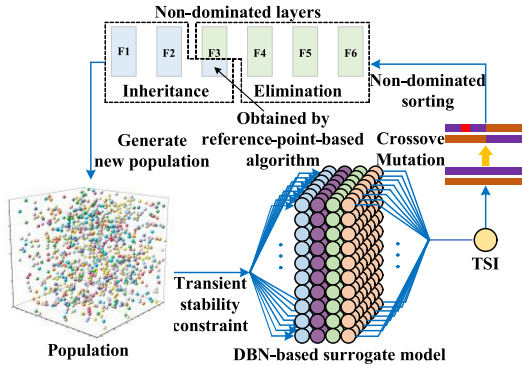


Fig. 2. Proposed surrogate model-assisted preventive control using NSGA-III algorithm.

algorithms, such as bat algorithm, firefly algorithm, and clonal selection algorithm, only handle single objective optimization and therefore not eligible for the multiobjective optimization in this article. To deal with that, the NSGA-III [25] is adopted, the advantages of which as compared to other multiobjective optimization algorithms have been proved. NSGA-III can find the multiobjective Pareto-optimal front. The proposed solution methodology for the surrogate-assisted preventive control is shown in Fig. 2. The optimization model aims to minimize the following four objectives simultaneously subject to equality and inequality constraints (27) and (28).

- 1) The total cost, including the costs of shifting the generator outputs and load shedding according to preventive control strategy; it is calculated using (22)–(26).
- 2) The index TSI

$$TSI_P = \begin{cases} TSI_B, & \mathcal{L}(\Lambda^{pre}) > TSI_B \\ \mathcal{L}(\Lambda^{pre}), & \mathcal{L}(\Lambda^{pre}) < TSI_B \end{cases} \quad (30)$$

$$\Lambda^{pre} = (\mathbf{P}_G^{pre}, \mathbf{V}_G, \mathbf{P}_{LS}^{pre}, \mathbf{P}_{LR}, \mathbf{Q}_L)^T \quad (31)$$

 TABLE I  
IMPLEMENTATION OF NSGA-III ALGORITHM

**Algorithm 1** NSGA-III ALGORITHM FOR OPTIMIZATION

**Input:** Transient unstable scenarios.

**Initialize:** Set the fluctuation ranges of the generator outputs and loads in the NSGA-III algorithm and randomly generate initial parent population  $NP_1$  of generator outputs and loads with size  $N$  that are within these ranges.

**Repeat:**

- 1 At the  $i$ th iteration, generate the offspring population  $NQ_i$  with size  $N$  by  $NP_i$  through crossover and mutation, where  $NP_i$  and  $NQ_i$  are combined to generate a population  $K_i$  with a size of  $2N$ .
- 2 Based on the power flow calculations and DBN-based surrogate model, the multi-objective values of solutions in  $K_i$  are obtained, where (22)–(26) and (30)–(31) are used.
- 3 Use non-dominated sorting to divide  $K_i$  into several non-dominated layers ( $F_1, F_2$ , etc.).
- 4 In sequence of the non-dominated layer, the solutions of each non-dominated layer are added to a new population  $NP_{i+1}$  until the size of  $NP_{i+1}$  is equal to  $N$ , or greater than  $N$  for the first time when adding solutions in non-dominated layer  $F_i$ . The solutions in the  $F_{i+1}$  and later layers will be eliminated.
- 5 The insufficient solutions of  $NP_{i+1}$  are selected from  $F_i$  through the following steps: *a.* Determining the reference points on the hyperplane; *b.* Adaptive normalization of population solutions; *c.* Associating each population solution with a reference point; *d.* Niche-preservation operation and *e.* fill all solutions of  $NP_{i+1}$ .

**Stop:** The preset number of iterations is reached.

**Output:** Optimal preventive control strategies ( $F_1$  of the last iteration).

where  $\mathbf{P}_G^{pre}$  represents the generator outputs given by the preventive control strategy;  $\mathbf{P}_{LS}^{pre}$  denotes the active power of loads that can be shed;  $\mathbf{P}_{LR}$  is the active power of remaining loads; except for  $\mathbf{P}_G^{pre}$  and  $\mathbf{P}_{LS}^{pre}$ , the other operational features are initial values, which are used together to form  $\Lambda^{pre}$ ;  $TSI_P$  larger than  $TSI_B$  is set to  $TSI_B$  instead of pursuing higher  $TSI_P$  to reduce control cost.

- 1) The third and fourth optimization objectives are to reduce the load ratio of generators and transmission lines. The average load ratio can be calculated using (23).

NSGA-III uses a set of well-spread reference points instead of crowding degree ranking to maintain population diversity as compared to the NSGA-II algorithm [25]. This leads to improved performance. At each iteration, the worst-case computational complexity of NSGA-III is the larger one of  $O(N^2 \log^{M-2} N)$  or  $O(N^2 M)$ , where  $M$  is the dimension of the objectives. Its implementations can be found in Table I.

In the NSGA-III algorithm, the TSI of the population is repeatedly calculated using the DBN-based transient stability predictor (the surrogate model) during the optimal preventive control strategy searching process. Assuming DBN has  $u$  layers, only  $u-1$  matrix multiplications and  $u-2$  activation function

calculations are needed to predict TSI. Thus, the surrogated-assisted model allows us to calculate TSI in a very efficient manner. In the meanwhile, DBN can input all solutions of a population in the form of a matrix and this enables us to get TSIs of all solutions through the same calculation process with a single TSI calculation. This greatly improves the convergence speed of the NSGA-III algorithm.

It is worth pointing out that only the optimal solution that meets the actual stability demands of the power system will be selected and the selection criteria are as follows.

- 1) Eliminate solutions that violate any individual constraint.
- 2) Transient stability is the primary objective of preventive control. Therefore, the solutions are selected according to the following aspects: (a) in most studies,  $TSI > 0$  generally means the system is transient stable, but in fact, when TSI is slightly higher than 0, the system is located at critical stability point. With the increase of simulation time, the power system is likely to change from the critical stability to instability. Thus, based on the conservatism principle of preventive control, a constant  $TSI_B$  that is greater than 0 is selected, (b) in control theory, there are definitions of stable, critical stable, and unstable. The definition of critical unstable is based on conservative control. For example, in [18], the boundary point between stable and critical stable is defined as  $TSI = 10$ . In this article, a sensitivity test, which investigates the relationship between  $TSI_B$  and  $A_{PC}$ , is performed. According to the results,  $TSI_B$  is selected as 60 to make  $A_{PC}$  reach 100%. At this time,  $TSI > TSI_B = 60$  means that when the predicted  $TSI > 60$ , the system is considered transient stable, while the predicted  $TSI < 60$  is considered unstable and preventive control should be performed to stabilize the system. Note that that in a special situation, such as heavy loading, the actual demand can be met by reducing the stability criterion, i.e.,  $TSI > TSI_B = 30$  can be considered; and (c) a too large TSI increase means a large cost. To this end, the dispatcher will comprehensively consider the TSI, the total cost, the load ratio of the generators, and transmission lines when determining the optimal preventive control strategy.

The detailed steps for the DBN-based surrogate model for preventive control are summarized in Fig. 3.

#### IV. CASE STUDY

In this section, the test system and data samples for training will be first shown. Then, the validity of the DBN for surrogate modeling will be demonstrated via simulations and comparison results with other alternatives. Finally, the integration of DBN results with the optimal model for generator redispatch and load shedding decision-making is analyzed.

##### A. Test Systems and Generation of Data Samples

The proposed method is tested on IEEE 39, 68, and 140-bus systems. First, data samples need to be generated. For three test systems, generator outputs and active power of the loads

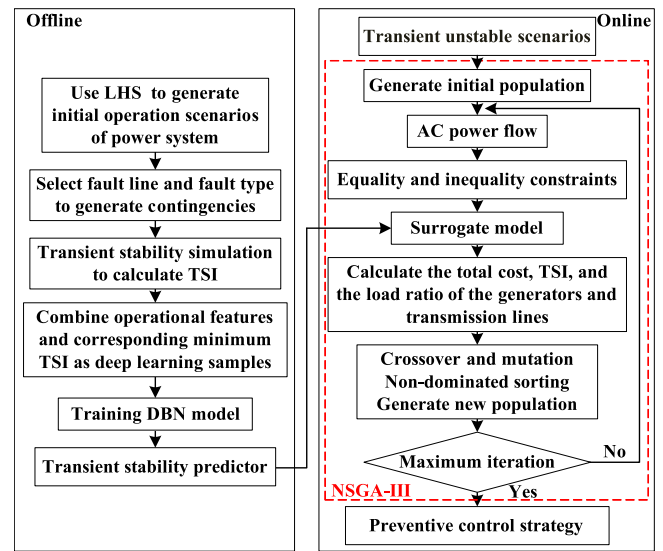


Fig. 3. Proposed DBN-based surrogate model for preventive control.

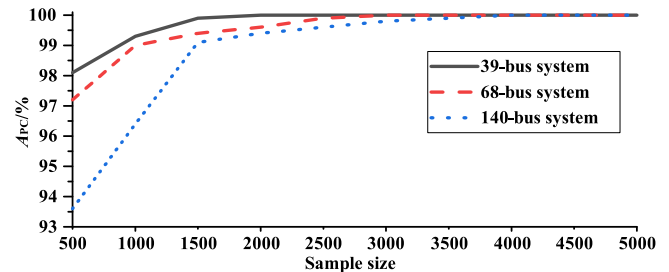


Fig. 4. Sensitivity analysis of the relationship between sample size and  $A_{PC}$ .

are assumed to fluctuate ranging from 80 to 120%; the voltage magnitudes of the generators are assumed to fluctuate in the range of 1.0–1.05, and the reactive power of the loads is derived from the active power to ensure the constant power factor. Considering the severity of the contingencies according to historical operational experience, 13, 19, and 31 three-phase short-circuit faults with fault duration of 0.1s are selected as contingencies for three test systems. Then, for three test systems, the sample sizes are set to be 2000, 3000, and 4000, respectively, according to the sensitivity analysis shown in Fig. 4, and the LHS is used to generate the corresponding operational scenarios. For each test system, operational features and contingencies are combined to form 26 000, 57 000, and 124 000 transient fault scenarios, respectively. The time-domain simulation period is 20 s. For each scenario, the vector of operational features combined with its corresponding minimum TSI is treated as one data sample. Note that for systems with larger scales and more complex structures, more samples are needed to cover diverse scenarios depending on the number of credible contingencies. This justifies the choice of sample size and this number can be increased if the system complexity increases and when more credible contingencies need to be considered.

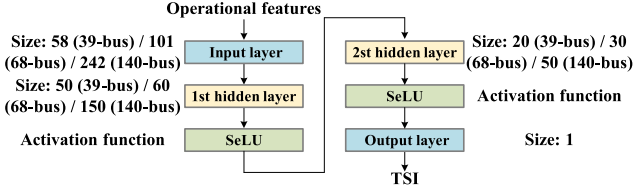


Fig. 5. Structure of the proposed DBN-based transient stability predictor.

TABLE II  
ACCURACY OF TRANSIENT STABILITY PREDICTOR

| Test system    | Prediction accuracy |              |           |
|----------------|---------------------|--------------|-----------|
|                | $TSI_B = 0$         | $TSI_B = 60$ | $A_{DBN}$ |
| 39-bus system  | 99.4%               | 100%         | 96.3%     |
| 68-bus system  | 98.9%               | 100%         | 93.8%     |
| 140-bus system | 99.0%               | 100%         | 93.9%     |

### B. Comparisons of Transient Stability Predictors

The samples are randomly divided into a training set and a test set, accounting for 80% and 20%, respectively. The training set is used to train the transient stability predictor while the test set is for validation. Then, (18) is used for data sample normalizations. The three predictors corresponding to three test systems have four layers with a number of neurons 58-50-20-1, 101-60-30-1, and 242-150-50-1, respectively, and batch size 5, epoch 100, initial learning rate 0.01, and activation function SeLU. The structure of the predictor is shown in Fig. 5. The prediction accuracies ( $A_{PC}$  and  $A_{DBN}$ ) for different test systems are shown in Table II. For  $A_{PC}$ , two  $TSI_B$  cases are considered and tested. It can be found that the proposed predictor achieves the best performance of preventive control for different systems, where the results for  $TSI_B = 60$  are more conservative than that for  $TSI_B = 0$ . This demonstrates the effectiveness of the proposed predictor for surrogate modeling. Also,  $TSI_B$  can be set according to actual needs by modifying (30) in the NSGA-III algorithm.

The ANN, SVM, convolutional neural network (CNN), and recurrent neural network (RNN) are also constructed to compare their performances with the proposed predictor. The first ANN consists of three layers, and the number of neurons is 10-50-1, i.e., ANN<sup>3</sup>. The second ANN consists of four layers, which adopt the same network structure and has the number of neurons the same as the predictor, i.e., ANN<sup>4</sup>. The optimizer, activation function, loss function, and data preprocessing method are the same as the proposed predictor. The SVM is constructed by scikit-learn, which is a machine learning package in Python. The CNN and RNN are constructed by Tensorflow. Specifically, one-dimensional CNN (1D CNN) is used, where the convolution window is 7, max pooling is used, and activation function is set to ReLU. RNN selects long short-term memory (LSTM) with dropout and ReLU. Note that both 1D CNN and LSTM have the same numbers of layers, iterations and optimizers as DBN. The prediction accuracies and losses for six methods in three

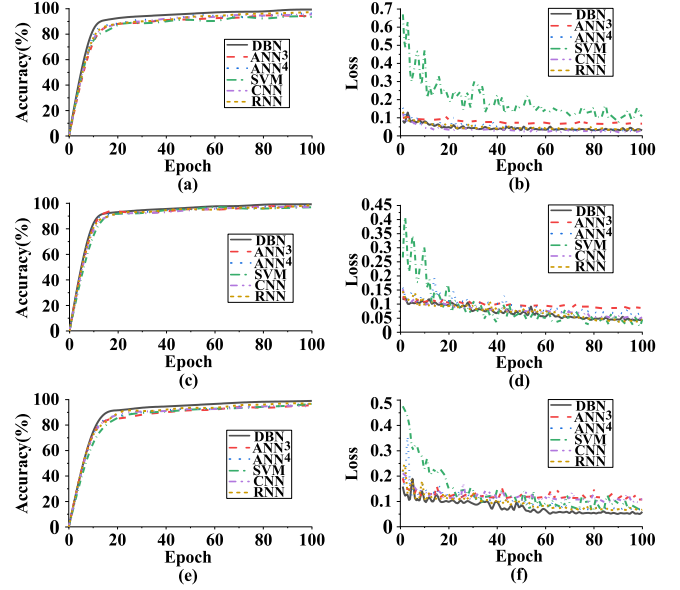


Fig. 6. Comparison results for DBN, ANN<sup>3</sup>, ANN<sup>4</sup>, SVM, CNN, and RNN. (a) and (b) Prediction accuracy and loss in the 39-bus system. (c) and (d) Prediction accuracy and loss in the 68-bus system. (e) and (f) Prediction accuracy and loss in the 140-bus system.

test systems are shown in Fig. 6, where the loss is characterized by the MSE of  $TSI_T$  and  $TSI_P$  after normalization. From Fig. 6, we can observe that DBN has the fastest convergence speed, minimal test set loss and the best prediction performance as compared to the other five alternatives. This is mainly due to the pretraining step that allows DBN to achieve multiple feature reconstructions from the input data. It also solves the problems of gradient disappearance and gradient explosion of deep neural networks, such as ANN<sup>4</sup>. It is worth noting that CNN is mainly used in image recognition, focusing on the local information of the image, while RNN is widely used in natural language processing. It mainly focuses on sequence data, such as time-series data. They are popular methods in their respective fields, but not suitable for the application here according to the test results.

### C. Preventive Control Without Load Shedding

As shown in Fig. 2, the DBN-based surrogate model is embedded in the iterative optimization process of the NSGA-III algorithm to search preventive control to avoid instability considering predefined contingencies. In order to determine the parameters of the NSGA-III algorithm, the population sizes are set to 100, 200, and 300, called  $S_{100}$ ,  $S_{200}$ , and  $S_{300}$  respectively, and their comparison results are shown in Table III. From Table III, we can see that  $S_{200}$  and  $S_{300}$  are the nondominated solutions as compared to that of  $S_{100}$ . Therefore, the population size needs to be larger than 100. It should be noted that the differences between the solutions of  $S_{200}$  and  $S_{300}$  are minor while the one with a smaller population size has a shorter iteration time. Therefore, the population size is set to 200. To further investigate the impacts of setting different numbers of iterations of NSGA-III,

TABLE III  
CHOICE OF NONDOMINATED DIFFERENT POPULATION SIZES

| Objectives                    | Population size |      |      |
|-------------------------------|-----------------|------|------|
|                               | 100             | 200  | 300  |
| TSI                           | 60              | 60   | 60   |
| Total cost                    | 1818            | 1146 | 1773 |
| Generators load ratio         | 62.5            | 62.4 | 61.6 |
| Transmission lines load ratio | 44.9            | 44.4 | 44.3 |

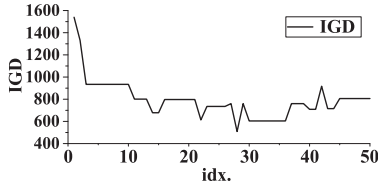


Fig. 7. Inverse generational distance at a population size of 200.

TABLE IV  
COMPARISON OF ONLINE COMPUTATIONAL EFFICIENCY

| Algorithm                            | Computation time |               |                |
|--------------------------------------|------------------|---------------|----------------|
|                                      | 39-bus system    | 68-bus system | 140-bus system |
| NSGA-III with DBN surrogate model    | 47s              | 49s           | 58s            |
| NSGA-III with time-domain simulation | >40min           | >1.5h         | >2.5h          |

the inverse generational distance (IGD) curve that can evaluate the convergence and distribution performance is used and its corresponding results with different iteration numbers are shown in Fig. 7. Since a good solution set corresponds to a small IGD, thus the number of iterations is set as 30 according to the results shown in Fig. 7. To make the reference points widely distributed on the entirely normalized hyperplane, the number of reference points is set to 10 000. The online computational efficiencies of the NSGA-III algorithm for the surrogate model-based and the original time-domain simulation-based methods for three systems are compared and the results are displayed in Table IV. It can be found that the use of the DBN-based surrogate model can significantly improve the computational efficiency of the NSGA-III algorithm, i.e., more than 30 times faster. It is also interesting to find that, unlike the physical-model-based TSC-OPF, the computing time of the proposed preventive control method only slightly increases with the increased scale of power systems. This is because system size has a negligible effect on some operations of the NSGA-III algorithm, such as non-dominated sorting that accounts for a large percentage of the total calculation time. The offline process includes time-domain simulations to generate training samples and surrogate model training (185.6, 189.2, and 198.7 s, respectively). The time of the online application is shown in Table IV. It should be noted that the offline process only needs to be performed once, unless the system has significant changes, such as topology changes beyond  $N - 1$ . Once the training is done, the proposed method only needs less than 1 min for preventive control.

Furthermore, the generator dispatch results obtained from the preventive control strategy for the three test systems are

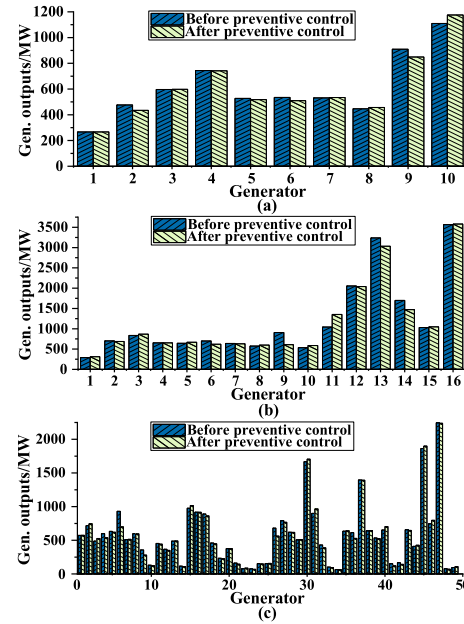


Fig. 8. Generator redispatch results without load shedding. (a) 39-Bus system. (b) 68-Bus system. (c) 140-Bus system.

TABLE V  
TOTAL COSTS OF 39-BUS SYSTEM

| Gen.      | Before control /MW | After control /MW | Output change /MW | Unit up cost/\$ | Unit down cost/\$ | Gen. cost/\$  | Total cost/\$ |
|-----------|--------------------|-------------------|-------------------|-----------------|-------------------|---------------|---------------|
| 1         | 266.7              | 266.5             | -0.2              | 8.78            | 3.42              | 0.68          | 1146          |
| <b>2</b>  | <b>476.4</b>       | <b>434.2</b>      | <b>-42.2</b>      | <b>8.81</b>     | <b>2.72</b>       | <b>114.78</b> |               |
| 3         | 596.9              | 597.7             | 0.8               | 9.81            | 7.54              | 7.85          |               |
| 4         | 743.9              | 742.1             | -1.8              | 8.42            | 4.97              | 8.95          |               |
| 5         | 527.1              | 518.2             | -8.9              | 10.75           | 3.53              | 31.42         |               |
| 6         | 534.9              | 509.8             | -25.1             | 13.40           | 3.27              | 82.08         |               |
| 7         | 531.1              | 534               | 2.9               | 8.76            | 8.16              | 25.4          |               |
| 8         | 446.4              | 456.5             | 10.1              | 10.17           | 8.21              | 102.72        |               |
| 9         | 909.8              | 848.8             | -61               | 7.96            | 4.20              | 256.2         |               |
| <b>10</b> | <b>1109.9</b>      | <b>1175.5</b>     | <b>65.6</b>       | <b>7.87</b>     | <b>3.36</b>       | <b>516.27</b> |               |

The significance of bold entities is to highlight numbers, which have the highest Gen. cost/\$.

shown in Fig. 8. It can be observed that the generator outputs slightly fluctuate before preventive control. This is because the NSGA-III algorithm can control the generation dispatch cost and reduce the adjustment range of generator outputs while maintaining system stability constraints. Setting  $C_U$  and  $C_D$ , according to (22)–(26), the total costs of 39, 68, and 140-bus systems are \$1146, \$4912, and \$6582, respectively. The detailed calculation processes of the total cost of the 39-bus system are shown in Table V. It can be observed that Gen. 10 has the maximum power output increase, i.e., 65.6 MW, while it has a lower  $C_U$ , indicating that preventive control tends to regulate the generator with a smaller cost. Similarly, Gen. 2 with lower  $C_D$  has a larger output decrease. After that, the transient stability for each contingency can be guaranteed; see Table VI for more details.

To validate the effectiveness of the proposed controls, the time-domain simulation results are displayed. The preventive



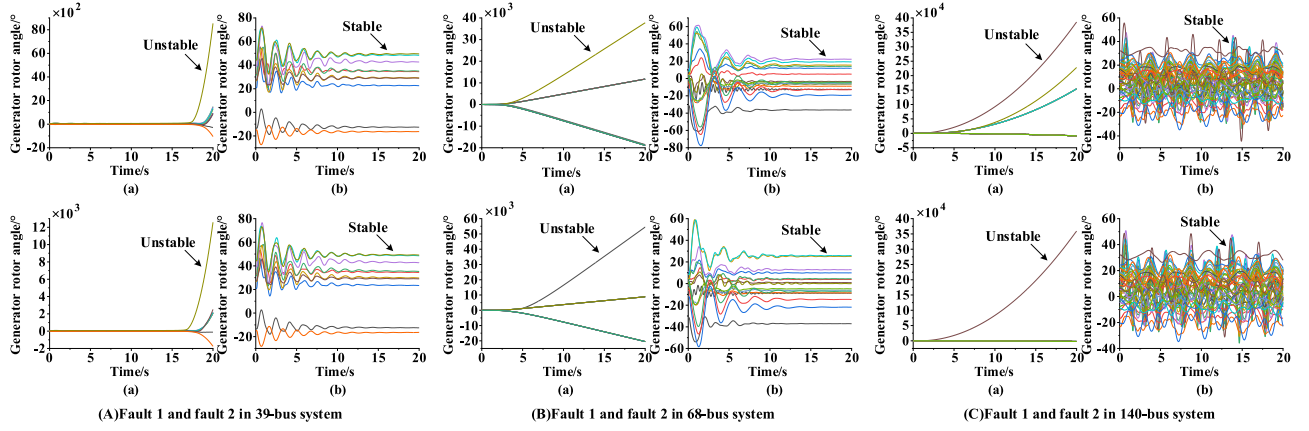


Fig. 9. Comparison of rotor angle trajectories before and after preventive control in three test systems.

TABLE VI  
COMPARISONS OF TSI BEFORE AND AFTER PREVENTIVE CONTROL

| Fault ID | 39-bus system  |               | 68-bus system  |               | 140-bus system |               |
|----------|----------------|---------------|----------------|---------------|----------------|---------------|
|          | Before control | After control | Before control | After control | Before control | After control |
| 1        | -92.8          | 69.1          | -98.7          | 72.0          | -99.8          | 75.0          |
| 2        | -95.1          | 69.2          | -99.0          | 70.3          | -99.8          | 75.2          |
| 3        | -97.5          | 70.1          | -98.9          | 78.5          | -99.8          | 74.8          |
| 4        | -11.9          | 69.5          | -96.5          | 74.4          | -99.8          | 70.4          |
| 5        | -87.5          | 68.7          | -96.1          | 72.2          | -99.8          | 62.0          |
| 6        |                |               |                |               | -99.7          | 70.0          |

control results for those faults that cause system instability are displayed in Table VI.

From Table VI, we can see that after performing the proposed preventive control, TSIs for all the contingencies are greater than  $TSI_B$ . For example, in the 39-bus system, the unstable cases, i.e., fault 1 ( $TSI = -92.8$ ) and fault 2 ( $TSI = -95.1$ ), are controlled to be stable with the value of index higher than 60. To achieve this, the traditional approach needs nearly an hour, while the proposed method only spends less than 1 min (see Table IV). The comparisons of rotor angle trajectories corresponding to the first two fault scenarios in Table VI before and after preventive control in three test systems are shown in Fig. 9. It is evident that all diverging rotor angle trajectories before preventive control have now been turned into stable cases, demonstrating the effectiveness of the proposed model. Furthermore, the load ratios of generators and transmission lines before and after preventive control are calculated using (23) and the results are shown in Table VII. It can be concluded that after preventive control, the load ratios of generators and transmission lines have been decreased significantly. Note that for each specific power networks, dispatchers can control the frequently overloaded devices to reduce the load ratio and ensure the safe operation of critical devices.

Finally, the Pareto-optimal front of TSI, total cost, and transmission line load ratio for the 39-bus system is shown in Fig. 10, where each point represents a solution of NSGA-III. The value of the point on each objective is mapped on the face of the

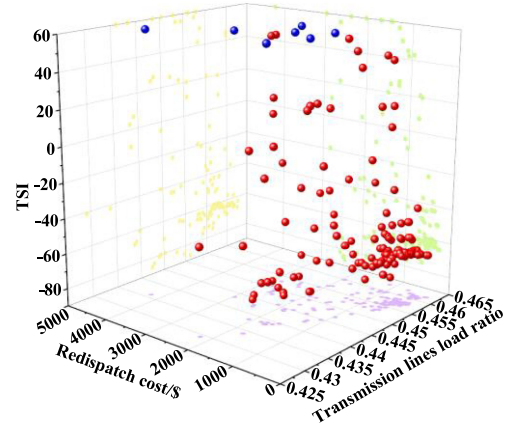


Fig. 10. Pareto-optimal front of the proposed method in the 39-bus system.

cube. It can be found that the TSIs of red points are less than 60, and therefore they will be discarded. All blue points are transient stability solutions. Note that a smaller transmission line load ratio corresponds to higher cost and dispatchers need to choose the best solution in the blue points according to the actual situation. When all constraints are met, the preventive control strategy with the lowest cost will be determined.

#### D. Preventive Control With Load Shedding

In some scenarios, such as heavy loading, generator redispatch may not be sufficient to guarantee system stability and load shedding is needed in the preventive control. Assuming that the three systems have 5, 6, and 10 loads that can be shed. Using the same scenarios as the previous case studies, the generator outputs and the loads which can be shed are used as the regulating variables in the NSGA-III algorithm, and the preventive control strategy is obtained. It is worth noting that the active power of the loads is always the input of DBN. When the preventive control without load shedding is performed,  $P_{LS}^{pre}$  represents the initial parameters. When the preventive control with load shedding is used,  $P_{LS}^{pre}$  is obtained by the preventive control strategy, namely

TABLE VII  
COMPARISON OF LOAD RATIO BEFORE AND AFTER PREVENTIVE CONTROL

| Load ratio (%)                  | 39-bus system  |               | 68-bus system  |               | 140-bus system |               |
|---------------------------------|----------------|---------------|----------------|---------------|----------------|---------------|
|                                 | Before control | After control | Before control | After control | Before control | After control |
| Generators                      | 62.7           | 62.4          | 60.9           | 60.5          | 57.0           | 55.9          |
| Transmission lines active power | 46.4           | 44.4          | 45.7           | 42.0          | 40.1           | 39.8          |

TABLE VIII  
COMPARISON OF OPTIMIZATION OBJECTIVES BEFORE AND AFTER PREVENTIVE CONTROL CONSIDERING LOAD SHEDDING

| Optimization objectives              | 39-bus system  |               | 68-bus system  |               | 140-bus system |               |
|--------------------------------------|----------------|---------------|----------------|---------------|----------------|---------------|
|                                      | Before control | After control | Before control | After control | Before control | After control |
| Minimum TSI                          | -97.5          | 66.8          | -99.0          | 65.2          | -99.8          | 56.5          |
| Load ratio of generators / %         | 62.7           | 61.6          | 60.9           | 52.8          | 57.0           | 56.3          |
| Load ratio of transmission lines / % | 46.4           | 44.1          | 45.7           | 32.3          | 40.1           | 38.6          |
| Cost / \$                            | 2327           |               | 14887          |               | 15325          |               |

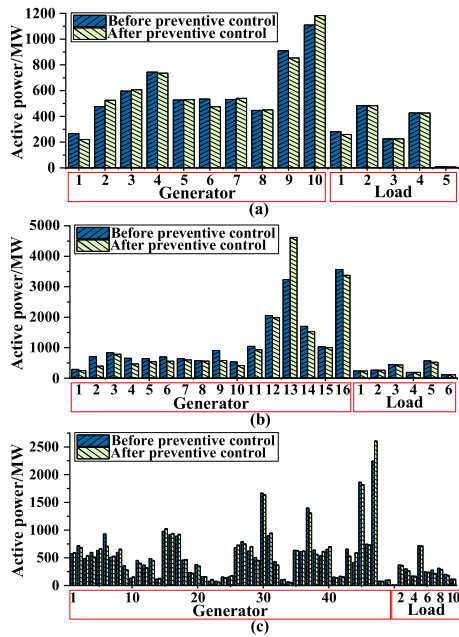


Fig. 11. Generator redispatch results considering load shedding. (a) 39-Bus system. (b) 68-Bus system. (c) 140-Bus system.

$P_{LS}^{pre} + P_{LR} = P_L$ . As a result, there is no dimensionality issue as compared to the previous case shown in Section IV-C. It is worth noting that this is also an advantage of the proposed method since the DBN model does not need to be retrained regardless of the location and quantity of load shedding.

The corresponding results are shown in Fig. 11, where the comparisons of TSI before and after preventive control are displayed in Table VIII. As shown in Table VIII, by generator dispatch and load shedding, a minimum TSI similar to that of using generator dispatch is obtained (see results in Tables VI and VII). In terms of load ratio, due to load shedding, it is significantly better than generator dispatch only. However, in terms of cost, the cost per MW of load shedding is about two times of increasing the generator outputs and five times of decreasing the generator outputs. Thus, the cost is significantly

higher than that of only generator dispatch. The results of preventive control considering load shedding show that the proposed method can effectively deal with different controlled variables and objectives.

#### E. Impacts and Countermeasures for DBN Mislabeling

When the proposed method is applied online, it is necessary to use DBN-based transient stability predictor to judge the risk of transient instability and decide whether to carry out transient stability preventive control. There are two cases of DBN mislabeling: classifying unstable scenario as stable one and classifying stable scenario as unstable one. For the first case, we can set a higher  $TSI_B$  to get conservative results depending on practical needs. As long as  $TSI_P$  is less than  $TSI_B$ , we can carry out preventive control to reduce the security risk with the increased cost. For the second case, an unnecessary preventive control will be implemented, yielding increased cost, which is the sum of costs for generation redispatch and load shedding. This has already been considered in the objective function (22) and the results are shown in Section IV-D. Therefore, in practical applications,  $TSI_B$  can be used to achieve the tradeoff between system safety and control cost.

#### F. Preventive Control Considering Topology Changes

Contingency analysis is typically studied for those credible ones that are well defined by the operators according to historical experiences. The focus of preventative control is usually used to stabilize the system if these contingencies lead to stability issues. Indeed, the system may undergo some extreme scenarios, yielding severe contingencies that are not included in the contingency list, such as those  $N-k$  contingencies. In this case, the emergent controls have to be used, which are out of the scope of this article. It is also worth noting that once these unstudied scenarios happen, they can be easily included in the training set and a new model can be obtained so that it can deal with it next time. By iterating the process, the model will become better and better.

In this article, some high probability and severe contingencies determined by the operator according to engineering experiences are selected to verify the proposed method. For the three test

TABLE IX

COMPARISON OF THE TSI BEFORE AND AFTER PREVENTIVE CONTROL CONSIDERING  $N$ -2 CONTINGENCIES

| N-2 Fault ID | 39-bus system  |               | 68-bus system  |               | 140-bus system |               |             |
|--------------|----------------|---------------|----------------|---------------|----------------|---------------|-------------|
|              | Before control | After control | Before control | After control | Before control | After control |             |
| LS           | 1              | <b>-99.8</b>  | <b>71.6</b>    | 78.8          | 86.8           | 63.5          | 77.9        |
|              | 2              | <b>-99.8</b>  | <b>71.7</b>    | <b>-84.0</b>  | <b>67.5</b>    | <b>-99.6</b>  | <b>75.4</b> |
|              | 3              |               |                | <b>-92.6</b>  | <b>86.8</b>    | <b>-99.6</b>  | <b>71.7</b> |
|              | 4              |               |                |               |                | <b>-99.8</b>  | <b>69.4</b> |
|              | 5              |               |                |               |                | 69.9          | 76.7        |
| GO           | 1              | <b>-99.4</b>  | <b>72.3</b>    | 79.6          | 86.5           | <b>-99.8</b>  | <b>72.3</b> |
|              | 2              | 74.5          | 72.7           | <b>-98.6</b>  | <b>75.0</b>    | <b>-99.6</b>  | <b>78.8</b> |
|              | 3              | <b>-99.5</b>  | <b>70.3</b>    | 77.7          | 86.8           | <b>-99.8</b>  | <b>66.7</b> |
|              | 4              |               |                | <b>-89.2</b>  | <b>54.4</b>    | <b>-99.6</b>  | <b>77.0</b> |
|              | 5              |               |                | 75.1          | 74.1           | <b>-99.6</b>  | <b>78.1</b> |
|              | 6              |               |                |               |                | 63.0          | 77.6        |
|              | 7              |               |                |               |                | 63.2          | 78.1        |
|              | 8              |               |                |               |                | 67.8          | 80.1        |
|              | 9              |               |                |               |                | 65.5          | 75.7        |
|              | 10             |               |                |               |                | <b>-94.9</b>  | <b>65.7</b> |

LS: line switching; GO: generation outage.

The significance of bold entities is to highlight numbers, which successfully achieves preventive control.

TABLE X

COMPARISON OF THE METHOD IN THIS ARTICLE AND THE METHOD IN [13]

| Objectives                    | Method   |      |
|-------------------------------|----------|------|
|                               | Proposed | [13] |
| TSI                           | 60       | 60   |
| Total cost                    | 1146     | 2361 |
| Generators load ratio         | 62.4     | 63.1 |
| Transmission lines load ratio | 44.4     | 46.7 |

systems, we have added 2, 3, and 5  $N$ -2 line switching contingencies and 3, 5, and 10  $N$ -2 generation outage contingencies, respectively. These cases are integrated with the previous samples to retrain the DBN. The TSI comparison results of  $N$ -2 contingencies before and after the preventive control are shown in Table IX. We can see that all  $N$ -2 contingencies are stable after preventive control, which proves that the proposed method is suitable for  $N$ - $k$  contingencies and topology changes.

### G. Comparison With Other Models

In [13], a transient stability preventive control model based on ANN and imperialist competitive algorithm (ICA) is proposed. The proposed method is compared with it on the 39-bus system, and the results are shown in Table X. Note that the same initial scenarios are selected and the ANN<sup>4</sup> is embedded in ICA. ICA selects the same parameters as the NSGA-III algorithm, and only generator redispatch is considered. Since ICA can only solve the single-objective optimization problem, the weighted sum of multiobjective is taken as the optimization objective of ICA. It can be seen from Table X that the preventive control results by our proposed method are much better than the single-objective evolutionary algorithm in [13]; see the total cost and generator load ratio.

## V. CONCLUSION

In this article, a new computing paradigm that integrates deep learning-enabled surrogate model and the multiobjective evolutionary algorithm was proposed for power system transient stability preventive control. The key idea was to strategically integrate the DBN-enabled surrogate model with the NSGA-III algorithm when solving the optimal preventive control strategy with various equality and inequality constraints. DBN allows users to develop the surrogate model for system transient stability assessment, different from the traditional time-consuming time-domain simulations. The surrogate model significantly reduced the computational complexity of the full TSC-OPF by simply performing direct deep learning-based transient stability verification in each iterative searching step for the optimal preventive control strategy. Comparative results on several IEEE test systems showed that the proposed method achieved significantly improved computational efficiency without loss of control performance. In our future work, the developed method will be tested using practical systems and data.

## REFERENCES

- [1] Y. Xu, W. Zhang, and M. Chow, "A distributed model-free controller for enhancing power system transient frequency stability," *IEEE Trans. Ind. Informat.*, vol. 15, no. 3, pp. 1361–1371, Mar. 2019.
- [2] S. Xia, S. Bu, and J. Hu, "Efficient transient stability analysis of electrical power system based on a spatially paralleled hybrid approach," *IEEE Trans. Ind. Informat.*, vol. 15, no. 3, pp. 1460–1473, Mar. 2019.
- [3] K. Sun, S. T. Lee, and P. Zhang, "An adaptive power system equivalent for real-time estimation of stability margin using phase-plane trajectories," *IEEE Trans. Power Syst.*, vol. 26, no. 2, pp. 915–923, May 2011.
- [4] T. L. Vu and K. Tristin, "Lyapunov functions family approach to transient stability assessment," *IEEE Trans. Power Syst.*, vol. 31, no. 2, pp. 1269–1277, Mar. 2016.
- [5] W. S. Rosenthal, A. M. Tartakovsky, and Z. Huang, "Ensemble Kalman filter for dynamic state estimation of power grids stochastically driven by time-correlated mechanical input power," *IEEE Trans. Power Syst.*, vol. 33, no. 4, pp. 3701–3710, Jul. 2018.
- [6] K. Hua *et al.*, "Fast unscented transformation-based transient stability margin estimation incorporating uncertainty of wind generation," *IEEE Trans. Sustain. Energy*, vol. 6, no. 4, pp. 1254–1262, Oct. 2015.
- [7] P. Bhui and N. Senroy, "Real-time prediction and control of transient stability using transient energy function," *IEEE Trans. Power Syst.*, vol. 32, no. 2, pp. 923–934, Mar. 2017.
- [8] J. Lv, M. Pawlak, and U. D. Annakkage, "Prediction of the transient stability boundary based on nonparametric additive modeling," *IEEE Trans. Power Syst.*, vol. 32, no. 6, pp. 4362–4369, Nov. 2017.
- [9] K. Sun, S. Likhate, and V. Vittal, "An online dynamic security assessment scheme using phasor measurements and decision trees," *IEEE Trans. Power Syst.*, vol. 22, no. 4, pp. 1935–1943, Nov. 2007.
- [10] J. Lv, M. Pawlak, and U. D. Annakkage, "Prediction of the transient stability boundary using the lasso," *IEEE Trans. Power Syst.*, vol. 28, no. 1, pp. 281–288, Feb. 2013.
- [11] F. R. Gomez, A. D. Rajapakse, and U. D. Annakkage, "Support vector machine-based algorithm for post-fault transient stability status prediction using synchronized measurements," *IEEE Trans. Power Syst.*, vol. 26, no. 3, pp. 1474–1483, Aug. 2011.
- [12] S. A. Siddiqui, K. Verma, and K. R. Niazi, "Real-time monitoring of post-fault scenario for determining generator coherency and transient stability through ANN," *IEEE Trans. Ind. Appl.*, vol. 54, no. 1, pp. 685–692, Jan./Feb. 2018.
- [13] R. Lajimi and T. Amraee, "A two stage model for rotor angle transient stability constrained optimal power flow," *Int. J. Elect. Power Energy Syst.*, vol. 76, pp. 82–89, 2016.
- [14] Q. Zhu *et al.*, "A deep end-to-end model for transient stability assessment with PMU data," *IEEE Access*, vol. 6, pp. 65474–65487, Oct. 2018.

- [15] A. Pizano-Martínez, C. R. Fuerte-Esquivel, and E. A. Zamora-Cárdenas, "Directional derivative-based transient stability-constrained optimal power flow," *IEEE Trans. Power Syst.*, vol. 32, no. 5, pp. 3415–3426, Sep. 2017.
- [16] Y. Xu, Z. Y. Dong, and J. Zhao, "Trajectory sensitivity analysis on the equivalent one-machine-infinite-bus of multi-machine systems for preventive transient stability control," *IET Gener. Transmiss. Distrib.*, vol. 9, no. 3, pp. 276–286, Feb. 2015.
- [17] Y. Xu, Z. Y. Dong, and L. Guan, "Preventive dynamic security control of power systems based on pattern discovery technique," *IEEE Trans. Power Syst.*, vol. 27, no. 3, pp. 1236–1244, Aug. 2012.
- [18] C. Liu *et al.*, "A systematic approach for dynamic security assessment and the corresponding preventive control scheme based on decision trees," *IEEE Trans. Power Syst.*, vol. 29, no. 2, pp. 717–730, Mar. 2014.
- [19] D. Lim, Y. Jin, and Y. Ong *et al.*, "Generalizing surrogate-assisted evolutionary computation," *IEEE Trans. Evol. Comput.*, vol. 14, no. 3, pp. 329–355, Jun. 2010.
- [20] F. Tian *et al.*, "A preventive transient stability control method based on support vector machine," *Electric. Power Syst. Res.*, vol. 170, pp. 286–293, May 2019.
- [21] Y. Liu *et al.*, "Intelligent online catastrophe assessment and preventive control via a stacked denoising autoencoder," *Neurocomputing*, vol. 3807, pp. 306–320, Mar. 2020.
- [22] R. T. F. A. King *et al.*, "Multi-contingency transient stability-constrained optimal power flow using multilayer feedforward neural networks," in *Proc. IEEE Can. Conf. Elect. Comput. Eng.*, Vancouver, BC, Canada, 2016, pp. 1–6.
- [23] Y. Zhou *et al.*, "Transient stability preventive control of power systems using chaotic particle swarm optimization combined with two-stage support vector machine," *Electric Power Syst. Res.*, vol. 155, pp. 111–120, Feb. 2018.
- [24] A. Costa and E. D. Buccio "Efficient parameter estimation for information retrieval using black-box optimization," *IEEE Trans. Knowl. Data Eng.*, vol. 30, no. 7, pp. 1240–1253, Jun. 2018.
- [25] K. Deb and H. Jain, "An evolutionary many-objective optimization algorithm using reference-point-based nondominated sorting approach, part I: Solving problems with box constraints," *IEEE Trans. Evol. Comput.*, vol. 18, no. 4, pp. 577–601, Aug. 2014.
- [26] T. Liu *et al.*, "A Bayesian learning based scheme for online dynamic security assessment and preventive control," *IEEE Trans. Power Syst.*, vol. 35, no. 5, pp. 4088–4099, Sep. 2020.



**Youbo Liu** (Member, IEEE) received the B.Sc., M.Sc., and Ph.D. degrees in electrical engineering from Sichuan University, Chengdu, China, in 2005, 2008, and 2011, respectively.

He is currently an Associate Professor in power system with the College of Electrical Engineering, Sichuan University. His research interests include machine learning in smart grid, energy-storage system applications, etc.

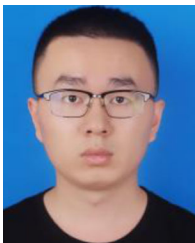


**Junbo Zhao** (Senior Member, IEEE) received the Ph.D. degree in electrical engineering from Virginia Tech, Blacksburg, VA, USA, in 2018.

He was a Research Assistant Professor with Virginia Tech from 2018 to 2019. He is currently an Assistant Professor with the Department of Electrical and Computer Engineering, Mississippi State University, Mississippi, MS, USA. He has published more than 100 peer-reviewed journal and conference papers, where more than 60 appear at IEEE Transactions. His

research interests include cyber-physical power system modeling, monitoring, uncertainty quantification, learning, dynamics, stability control, and security with DERs.

Dr. Zhao is currently serving as the Chair of IEEE Task Force on Power System Dynamic State and Parameter Estimation and IEEE Task Force on Cyber-Physical Interdependency for Power System Operation and Control, Co-Chair of the IEEE Working Group on Power System Static and Dynamic State Estimation. He also serves as an Editor for IEEE TRANSACTIONS ON POWER SYSTEMS, IEEE TRANSACTIONS ON SMART GRID, and IEEE POWER ENGINEERING LETTERS, Associate Editor for *International Journal of Electrical Power & Energy Systems* and *Journal of Modern Power System and Clean Energy*, and Subject Editor for *IET Generation, Transmission & Distribution* and *CSEE Journal of Power and Energy Systems*. He received the IEEE PES Outstanding Engineer Award in 2020 for contributions to the PES MS chapter.



**Tong Su** received the B.Sc. degree in 2019 from Sichuan University, Chengdu, China, where he is currently working toward the M.Sc. degree, both in electrical engineering.

His research interests include power system machine learning and transient stability analysis.



**Junyong Liu** (Member, IEEE) received the Ph.D. degree in electrical engineering from Brunel University, London, U.K., in 1998.

He is currently a Professor with the College of Electrical Engineering, Sichuan University, Chengdu, China. His research interests include power system planning, operation, and electricity marking.